p-ADIC HAHN SERIES WITH SPARSE SUPPORT

by

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Contents

1. Introduction

Let p be a prime number.

1.1. p-adic transcendence via p-adic Hahn series. — A p-adic Hahn series is a generalized formal power series of the form $f = \sum_{q\in\mathbf{Q}}[f(q)]p^q$, where $f(q) \in \overline{\mathbf{F}}_p$ and $[\cdot]$ is the Teichmüller lift, whose support $\mathrm{Supp}(f) = \{q \in \mathbf{Q} \mid f(q) \neq 0\}$ is a well-ordered subset of $\mathbf{Q}$. Krull, Lampert, and Poonen showed that the set of p-adic Hahn series, which we denote by $\mathbf{L}_p$, forms the spherical completion of $\overline{\mathbf{Q}}_p$, the field of p-adic algebraic numbers. In particular, $\mathbf{L}_p$ is algebraically closed and complete with respect to the p-adic valuation given by $f \mapsto \min \mathrm{Supp}(f)$.

The field $\mathbf{L}_p$ provides a natural setting in which to study transcendental number theory over p-adic fields. More precisely, a fundamental question arises:

Question 1.1. — *Given a p-adic Hahn series $f \in \mathbf{L}_p$, how can one determine whether f is algebraic over $\mathbf{Q}_p$?*

Although this question remains open in general, several necessary conditions for a p-adic Hahn series to be algebraic over $\mathbf{Q}_p$ are known:

1. In [Lam86] and [Poo93], Lampert and Poonen proved that if $f \in \mathbf{L}_p$ is algebraic over $\mathbf{Q}_p$, then
   (a) there exists an integer T such that $\mathrm{Supp}(f) \subset \frac{1}{T}\mathbf{Z}[1/p]$;
   (b) there exists a finite extension $\mathbf{F}_q$ of $\mathbf{F}_p$ such that $\{f(q)\}_{q\in\mathbf{Q}} \subset \mathbf{F}_q$.
   These conditions are also studied quantitatively in [WY24].
2. In [Lam86], Lampert also proved that if $f \in \mathbf{L}_p$ is algebraic over $\mathbf{Q}_p$, then the accumulation points of $\mathrm{Supp}(f)$ are rational numbers.

3. In [Ked01] and [Ked17], Kedlaya gives a necessary and sufficient condition for an equal-characteristic Hahn series in $\mathbf{L}_p^\flat := \overline{\mathbf{F}}_p(\!(t^{\mathbf{Q}})\!)$ to lie in the algebraic closure $\overline{\mathbf{F}}_p(\!(t)\!)^{\mathrm{alg}}$ of $\overline{\mathbf{F}}_p(\!(t)\!)$, phrased in the language of automata theory. As an application, Kedlaya uses Witt vectors to lift this result to the p-adic case: he shows that the field $\mathbf{C}_p := \widehat{\overline{\mathbf{Q}}}_p$ of p-adic complex numbers, when viewed as a subfield of $\mathbf{L}_p$, coincides with the completion of the set $\Theta\left(\overline{\mathbf{F}}_p(\!(t)\!)^{\mathrm{alg},\wedge}\right)$, where $\overline{\mathbf{F}}_p(\!(t)\!)^{\mathrm{alg},\wedge}$ is the t-adic completion of $\overline{\mathbf{F}}_p(\!(t)\!)^{\mathrm{alg}}$ and $\Theta\colon \overline{\mathbf{F}}_p(\!(t^{\mathbf{Q}})\!) \longrightarrow \mathbf{L}_p$ is the map $\sum_{q\in\mathbf{Q}} f(q)t^q \longmapsto \sum_{q\in\mathbf{Q}}[f(q)]p^q$.

Intuitively, Kedlaya's result indicates that the algebraicity of equal-characteristic Hahn series in $\overline{\mathbf{F}}_p(\!(t^{\mathbf{Q}})\!)$ and the algebraicity of p-adic Hahn series in $\mathbf{L}_p$ are related to some extent via the map Θ, which leads to the following questions:

qu:29274 ***Question 1.2.*** —

1. *Suppose that $f \in \mathbf{L}_p^\flat$ is algebraic over $\mathbf{F}_p(\!(t)\!)$. Is $\Theta(f)$ algebraic over $\mathbf{Q}_p$?*
2. *Suppose that $f \in \mathbf{L}_p$ is algebraic over $\mathbf{Q}_p$. Is $\Theta^{-1}(f)$ algebraic over $\mathbf{F}_p(\!(t)\!)$?*

In [WY25], we gave a negative answer to the first question by showing that the p-adic Hahn series $\mathfrak{A} := \sum_{k=1}^{\infty} p^{-1/p^k}$ is transcendental over $\mathbf{Q}_p$, whereas its preimage under Θ is a root of the polynomial $X^p - X - t^{-1}$ over $\mathbf{F}_p(\!(t)\!)$.

1.2. Sparseness and main theorem. — The key observation in [WY25] is that the support of $\mathfrak{A}$ is "sparse", in the sense that for any non-zero polynomial $P(X) \in \mathbf{Q}_p[X]$, the multinomial expansion of $P(\mathfrak{A})$ contains orphaned terms that cannot be cancelled by other terms, which forces $P(\mathfrak{A})$ to be nonzero. In this article, we generalize this idea to the following combinatorial definitions:

Definition 1.3. —

1. *Any rational number q can be uniquely written in the form $q = w + \sum_{i=1}^{\infty} q_i \cdot p^{-i}$, $q_i \in \{0, 1, \dots, p-1\}$, with $q_i \neq p-1$ for infinitely many i. We call $\mathfrak{N}_p(q) := \sum_{i=1}^{\infty} q_i \in \mathbf{N} \cup \{\infty\}$ the p-**digit sum** of q.*
2. *For any set S of rational numbers, define the **dominant** p-**digit sum** of S to be*
$$\mathrm{dom}_p(S) := \sup\{\mathfrak{N}_p(q) | q \in S\} \in \mathbf{N} \cup \{\infty\},$$
*and define the p-**digit dominant part** of S to be*
$$\mathrm{Dom}_p(S) := \{q \in S | \mathfrak{N}_p(q) = \mathrm{dom}_p(S)\}.$$

def:15658 ***Definition 1.4.*** — *A set $S \subset [0,1) \cap \mathbf{Q}$ is **sparse** if:*

1. *$\mathrm{dom}_p(S)$ is finite;*
2. *for infinitely many integers $n \geq 1$, there exist n elements $d_1, \cdots, d_n \in \mathrm{Dom}_p(S)$ such that*
   (a) *there is no carry when adding $d_1, \cdots, d_n$ together;*
   (b) *if $e_1, \cdots, e_n \in S$ satisfy that $d_1 + \cdots + d_n$ and $e_1 + \cdots + e_n$ differ by an integer, then up to a permutation of $\{1, 2, \cdots, n\}$, $d_i = e_i$ for every $i = 1, 2, \cdots, n$.*

Remark 1.5. — *We give some comments on Definition 1.4:*

1. *The first condition is closely related to Kedlaya's criterion for the algebraicity of equal-characteristic Hahn series: he shows that if $f \in \mathbf{L}_p^\flat$ is algebraic over $\overline{\mathbf{F}}_p(\!(t)\!)$, then $\mathrm{dom}_p(-T \cdot \mathrm{Supp}(f))$ is finite for some integer $T \geq 1$.*
2. *Condition (2a) indicates that the addition $d_1 + \cdots + d_n$, no matter how one parenthesizes it, behaves like the addition in the free commutative monoid $\bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.*
3. *Condition (2b) is a combinatorial rigidity condition asserting that the chosen carry-free sum admits a unique decomposition modulo $\mathbf{Z}$. This produces the orphaned exponents in the multinomial expansion, which is the key mechanism behind the transcendence proof.*

The following example illustrates a typical situation in which the sparseness condition is satisfied:

eg:61947 ***Example 1.6*** **(cf. Example 3.7).** — *Let $A_1, A_2, \cdots$ be a family of pair-wise disjoint nonempty subsets of $\mathbf{Z}_{\geq 1}$ such that $\sup_i |A_i| < \infty$, and this supremum is attained by infinitely many i. Then the set*

$$\left\{ \sum_{r \in A_i} p^{-r} \,\middle|\, i = 1, 2, \cdots \right\}$$

is sparse.

The main theorem of this article is the following:

thm:main ***Theorem 1.7*** **(cf. Theorem 5.1).** — *Let $f \in \mathbf{L}_p$ be a p-adic Hahn series such that there exists an integer $T \geq 1$ for which $-T \cdot \mathrm{Supp}(f)$ admits a sparse set $W \neq \{0\}$ of representatives modulo $\mathbf{Z}$. Then f is transcendental over $\breve{\mathbf{Q}}_p$, and hence over $\mathbf{Q}_p$.*

Remark 1.8. — *The sparseness condition does not involve the coefficients of f, so it is natural that it does not distinguish algebraicity over $\mathbf{Q}_p$ from algebraicity over $\breve{\mathbf{Q}}_p$.*

1.3. p-adic analogue of a result of Huang and Ştefănescu. — As mentioned in [Ked01, Section 1], Kedlaya's criterion for the $\overline{\mathbf{F}}_p$-algebraicity of Hahn series in $\mathbf{L}_p^\flat$ is motivated by the following result of Huang, which was independently discovered by Ştefănescu:

thm:11165 ***Theorem 1.9*** **(cf. [Hua68; Şte83]).** — *Let $f = \sum_{i=1}^{\infty} f(i) \cdot t^{-1/p^i} \in \mathbf{L}_p^\flat$ be a Hahn series. Then the following are equivalent:*

1. *the series f is algebraic over $\overline{\mathbf{F}}_p(\!(t)\!)$;*
2. *the series f is algebraic over $\mathbf{F}_p(\!(t)\!)$;*
3. *the sequence $\{f(i)\}_{i \geq 1}$ is eventually periodic.*

With the help of Theorem 1.7 (as well as Example 1.6), we prove the following p-adic analogue of Theorem 1.9:

thm:mtb ***Theorem 1.10*** **(cf. Corollary 5.4).** — *Let $f = \sum_{i=1}^{\infty} [f(i)] \cdot p^{-1/p^i} \in \mathbf{L}_p$ be a p-adic Hahn series. Then the following are equivalent:*

1. *the series f is algebraic over $\mathbf{Q}_p$;*
2. *the series f is algebraic over $\breve{\mathbf{Q}}_p$;*
3. *$f(i) = 0$ for all but finitely many i.*

Remark 1.11. — *Theorem 1.10 yields infinitely many counter-examples to the first question in Question 1.2: any Hahn series $f = \sum_{i=1}^{\infty} f(i) \cdot t^{-1/p^i} \in \mathbf{L}_p^\flat$ for which $f(i) \neq 0$ for infinitely many i and the sequence $\{f(i)\}_{i \geq 1}$ is eventually periodic is algebraic over $\mathbf{F}_p(\!(t)\!)$ by Theorem 1.9, while $\Theta(f)$ is transcendental over $\mathbf{Q}_p$ by Theorem 1.7.*

Acknowledgements. —

2. Preliminaries on Hahn series

To keep this article self-contained, we briefly recall some basic facts about Hahn series.

def:61947 ***Definition 2.1*** **([Poo93, Section 3]).** — *Let R be a commutative ring and G be an ordered group.*

1. *For any $f \in \mathrm{Hom}_{\mathrm{Set}}(G, R)$, we define the **support** of f to be*

$$\mathrm{Supp}(f) = \{g \in G \colon f(g) \neq 0\}.$$

2. *Define the set of* ***Hahn series*** *over R with value group G to be*

$$R(\!(G)\!) := \{f \in \mathrm{Hom}_{\mathrm{Set}}(G, R) \colon \ \mathrm{Supp}(f) \text{ is well-ordered}\}.$$

By introducing a formal variable t, elements in $R(\!(G)\!)$ will also be written as $\sum_{g\in G} r_g t^g$, where $r_g \in R$ for all $g \in G$.

prop:40704 ***Proposition 2.2*** **([Poo93, Lemma 1, Corollary 2]).** — *Let R be a commutative ring and G be an ordered group.*

1. *With identity $1 \cdot t^0$ and addition as well as multiplication given by*

$$\sum_{g\in G} a_g t^g + \sum_{g\in G} b_g t^g := \sum_{g\in G} (a_g + b_g) t^g, \ \sum_{g\in G} a_g t^g \cdot \sum_{g\in G} b_g t^g := \sum_{g\in G} \left(\sum_{h\in G} a_h b_{g-h} \right) t^g,$$

$R(\!(G)\!)$ forms a commutative ring.

2. *If R is a field, then so is $R(\!(G)\!)$. Moreover, with the map*

$$v \colon R(\!(G)\!) \longrightarrow G \cup \{\infty\}, \ f \longmapsto \begin{cases} \min \mathrm{Supp}(f), & \text{if } f \neq 0 \\ \infty, & \text{if } f = 0 \end{cases},$$

$R(\!(G)\!)$ becomes a valued field with value group G and residue field R.

Since $\operatorname{char} R(\!(G)\!) = \operatorname{char} R$, we call $R(\!(G)\!)$ the **equal-characteristic field of Hahn series** over R with value group G, also denoted by $R(\!(t^G)\!)$ with respect to the formal variable t.

prop:49318 ***Proposition 2.3*** **([Poo93, Proposition 3, Corollary 3, Proposition 5])**

Let k be a perfect field of characteristic p and G be an ordered group containing $\mathbb{Z}$ as a subgroup. Besides that, let

$$\mathcal{N} := \left\{ \sum_{g\in G} r_g t^g \in W(k)(\!(t^G)\!) \colon \ \text{for every } g \in G, \ \sum_{n\in\mathbb{Z}} r_{g+n} p^n = 0 \right\},$$

where $W(k)$ is the ring of Witt vectors of k. We call elements in $\mathcal{N}$ the ***null series*** *of $W(k)(\!(t^G)\!)$. Then*

1. *$\mathcal{N}$ is a maximal ideal of $W(k)(\!(t^G)\!)$, which makes $W(k)(\!(p^G)\!) := W(k)(\!(t^G)\!)/\mathcal{N}$ a field[(1)], called the* ***p-adic field of Hahn series****.*
2. *Every element in $W(k)(\!(p^G)\!)$ can be uniquely written as*

$$\sum_{g\in G} [r_g] p^g,$$

where $r_g \in k$ for all $g \in G$ and $[\cdot] \colon k \longrightarrow W(k)$ is the Teichmüller lift. We call this the ***standard expansion*** *of the element.*

3. *For $f = \sum_{g\in G} [r_g] p^g$, define the* ***support*** *of f to be*

$$\mathrm{Supp}(f) = \{g \in G \colon r_g \neq 0\}.$$

Then the map

$$v \colon W(k)(\!(G)\!)/\mathcal{N} \longrightarrow G \cup \{\infty\}, \ f \mapsto \begin{cases} \min \mathrm{Supp}(f), & \text{if } f \neq 0 \\ \infty, & \text{if } f = 0 \end{cases}$$

makes $W(k)(\!(G)\!)/\mathcal{N}$ a mixed-characteristic valued field with value group G and residue field k.

The most fundamental property of the field of Hahn series is the following:

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(1) Informally, $W(k)(\!(p^G)\!)$ is obtained by replacing the formal variable t in elements of $W(k)(\!(t^G)\!)$ by the prime p.

***Theorem 2.4* (cf. [Poo93, Theorem 1, Corollary 4, Corollary 6])**

Let F be an equal-characteristic (resp. mixed-characteristic) valued field with divisible value group G and algebraically closed residue field k. Then the equal-characteristic (resp. p-adic) field of Hahn series $k(\!(t^G)\!)$ (resp. $W(k)(\!(p^G)\!)$) is the unique (up to isomorphism of valued fields) minimal spherically complete extension of F. Moreover, it is algebraically closed and complete.

The following are the fields of Hahn series used in this article:

eg:44541 ***Example 2.5.*** — *Let $F = \overline{\mathbf{F}}_p(\!(t)\!)$ (resp. $\breve{\mathbf{Q}}_p = W(\overline{\mathbf{F}}_p)[p^{-1}]$), which has value group $\mathbf{Q}$ and residue field $\overline{\mathbf{F}}_p$. Then the field of equal-characteristic (resp. p-adic) Hahn series $\mathbf{L}_p^\flat := \overline{\mathbf{F}}_p(\!(t^{\mathbf{Q}})\!)$ (resp. $\mathbf{L}_p := \mathscr{O}_{\breve{\mathbf{Q}}_p}(\!(p^{\mathbf{Q}})\!) = W(\overline{\mathbf{F}}_p)(\!(p^{\mathbf{Q}})\!)$) is the spherical completion of F with the same residue field and value group, and is algebraically closed and complete.*

Remark 2.6. — *For brevity, we call $\mathscr{O}_{\breve{\mathbf{Q}}_p}(\!(p^{\mathbf{Q}})\!)$ the p-adic Hahn series without specifying the residue field and value group.*

3. Sparseness and (c, n)-sparseness

To give a rigorous and workable formulation of the sparseness condition in Definition 1.4, we introduce several combinatorial notions and auxiliary functions related to base-p expansions of rational numbers in $[0, 1)$.

Definition 3.1. —

1. *For $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let $\Sigma(\underline{d}) := \sum_{i=1}^{\infty} d_i \in \mathbf{N}$ and $\|\underline{d}\| := \sum_{i=1}^{\infty} d_i p^{-i} \in \mathbf{Q}_{\geq 0}$.*
2. *Let $\mathbb{P} := \bigoplus_{\mathbf{Z}_{\geq 1}} \{0, 1, \cdots, p-1\} \subsetneq \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.*

Lemma 3.2. — *For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, there exists a unique element $\tau(\underline{d}) \in \mathbb{P}$ such that $\|\underline{d}\| - \|\tau(\underline{d})\| \in \mathbf{Z}$.*

Proof. — If one writes $\|\underline{d}\|$ as a decimal expansion(2) in base p:

$$\|\underline{d}\| = w.d_1 \cdots d_n \cdots,$$

where $w \in \mathbf{Z}$ and $d_i \in \{0, 1, \cdots, p-1\}$ for any i, then $\tau(\underline{d}) = 0.d_1 \cdots d_n \cdots$. The uniqueness is trivial. □

We collect several properties of these concepts in the following lemma:

lem:18813 ***Lemma 3.3.*** — *Let $\underline{d}, \underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.*

it:28363 1. *The maps $\Sigma(\cdot)$ and $\|\cdot\|$ are additive. Moreover, $\|\cdot\|$ is injective when restricted to $\mathbb{P}$.*
it:52935 2. *One has $\tau(\underline{d}) = \tau(\underline{e})$ if and only if $\|\underline{d}\| - \|\underline{e}\| \in \mathbf{Z}$.*
3. *$\underline{d} \in \mathbb{P}$ if and only if $\tau(\underline{d}) = \underline{d}$.*
lem:17816 4. *One has $\Sigma(\tau(\underline{d})) \leq \Sigma(\underline{d})$, with equality if and only if $\underline{d} \in \mathbb{P}$.*

Proof. — The first three statements are straightforward. We only prove the last statement.

For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let

$$N(\underline{d}) := \begin{cases} 0, & \text{if } \underline{d} \in \mathbb{P}; \\ \max\{i | d_i \geq p\}, & \text{otherwise.} \end{cases}$$

By a descent argument on $N(\underline{d})$, it suffices to show that if $N(\underline{d}) \geq 1$, then there exists $\underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$ such that $\|\underline{d}\| - \|e\| \in \mathbf{Z}$, $\Sigma(\underline{e}) < \Sigma(\underline{d})$, and $N(\underline{e}) < N(\underline{d})$.

(2) We do not allow infinite strings of $(p-1)$ in the decimal expansion of $\|\underline{d}\|$ to ensure the uniqueness of $\tau(\underline{d})$.

Writing $d_{N(\underline{d})} = p{\cdot}r{+}s$ with $r \in \mathbf{N}$ and $s \in \{0, 1, \cdots, p{-}1\}$, we take $\underline{e} := (d_0, \cdots, d_{N(\underline{d})-2}, d_{N(\underline{d})-1} + r, s, 0, \cdots)$. Then

$$\Sigma(\underline{d}) - \Sigma(\underline{e}) = d_{N(\underline{d})} - r - s = (p-1) \cdot r > 0$$

and $\|\underline{d}\| - \|\underline{e}\| = 0$. The result follows. □

Remark 3.4. — *For a rational number q in $[0, 1)$ with a finite-length decimal expansion in base p, the preimage $\underline{q}$ of q under the map $\|\cdot\|$ (which must exist) extracts the digits of q in base p, and $\Sigma(\underline{q})$ is the p-digit sum of q.*

def:38144 ***Definition 3.5.*** — *Let p be a prime number. Let $c, n \geq 1$ be integers. A subset $S \subset \mathbb{P}$ is (c, n)-**sparse** if*

1. *there exists $c \geq 1$ such that $\Sigma(\underline{d}) \leq c$ for every $\underline{d} \in S$;*
2. *there exist n (not necessarily distinct) elements $\underline{d}^{(1)}, \underline{d}^{(2)}, \cdots, \underline{d}^{(n)} \in S$ such that*
   (a) *$\Sigma\left(\underline{d}^{(i)}\right) = c$ for $i = 1, 2, \cdots, n$ and $\sum_{i=1}^{n} d_j^{(i)} < p$ for $j \in \mathbf{N}$ (that is, $\sum_{i=1}^{n} \underline{d}^{(i)} \in \mathbb{P}$);*
   (b) *if $\underline{e}^{(1)}, \underline{e}^{(2)}, \cdots, \underline{e}^{(n)} \in S$ satisfy $\left\|\sum_{i=1}^{n} \underline{d}^{(i)}\right\| - \left\|\sum_{i=1}^{n} \underline{e}^{(i)}\right\| \in \mathbf{Z}$, then up to a permutation of $\{1, 2, \cdots, n\}$, $\underline{d}^{(i)} = \underline{e}^{(i)}$ for every $i = 1, 2, \cdots, n$.*

The following lemma reformulates the sparseness condition in Definition 1.4 in terms of the (c, n)-sparseness condition in Definition 3.5.

lem:54409 ***Lemma 3.6.*** — *A set $W \neq \{0\}$ is sparse in the sense of Definition 1.4 if and only if there exists a subset S of $\mathbb{P}$ such that $W = \|S\|$ and there exists an integer $c \geq 1$ such that S is (c, n)-sparse for infinitely many integers $n \geq 1$.*

Proof. — Suppose that $W \subset [0, 1) \cap \mathbf{Q}$ is sparse. Then the p-digit sum of every element in W is finite and bounded by $\mathrm{dom}_p(W)$. Thus we may take

$$S := \left\{ (q_i)_{i \in \mathbf{Z}_{\geq 1}} | q = w + \sum_{i=0}^{\infty} q_i p^{-i} \in W,\ q_i \in \{0, \cdots, p-1\} \right\}.$$

Then S is $(\mathrm{dom}_p(W), n)$-sparse for infinitely many integers $n \geq 1$.

The converse direction follows by a similar argument. □

We give a concrete example of the sparse set, which is the prototype of the situation in which the sparseness condition is satisfied:

eg:new ***Example 3.7.*** — *Let $\underline{A} := (A_i)_{i \geq 1}$ be a family of pair-wise disjoint nonempty subsets of $\mathbf{Z}_{\geq 1}$ such that $\sup_i |A_i| < \infty$ and this supremum is attained by infinitely many i. Then the set*

$$M(\underline{A}) := \left\{ \sum_{r \in A_i} p^{-r} \middle| i = 1, 2, \cdots \right\} \subset [0, 1) \cap \mathbf{Q}$$

is sparse.

Proof. — For any i, one has $\mathfrak{N}_p(\sum_{r \in A_i} p^{-r}) = |A_i|$, and consequently $\mathrm{dom}_p(M(\underline{A})) = \sup_i |A_i| < \infty$. The infinity of attainment of the supremum implies that $\mathrm{Dom}_p(M(\underline{A}))$ is an infinite set:

$$\mathrm{Dom}_p(M(\underline{A})) = \left\{ \mathbf{d}_j := \sum_{r \in A_{i_j}} p^{-r} \middle| j = 1, 2, \cdots \right\},$$

with $\mathbf{d}_k \neq \mathbf{d}_l$ for any $k \neq l$.

Take $e_1, \cdots, e_n \in M(\underline{A})$ such that $\sum_{j=1}^{n} e_j - \sum_{j=1}^{n} \mathbf{d}_j \in \mathbf{Z}$. Since $M(\underline{A}) \subset \|\mathbb{P}\|$, we may write $e_j = \left\|\underline{e}^{(j)}\right\|$ and $\mathbf{d}_j = \left\|\underline{\mathbf{d}}^{(j)}\right\|$ with $\underline{e}^{(j)}, \underline{\mathbf{d}}^{(j)} \in \mathbb{P}$ for every $j = 1, \cdots, n$. Then one has

$\sum_{j=1}^{n} \mathbf{d}_j = \sum_{r\in\bigsqcup_{j=1}^{n} A_{i_j}} p^{-r}$, and consequently $\sum_{j=1}^{n} \underline{\mathbf{d}}^{(j)} \in \mathbb{P}$. In particular,

$$n \cdot \mathrm{dom}_p(M(\underline{A})) = \mathfrak{N}_p\left(\sum_{r\in\bigsqcup_{j=1}^{n} A_{i_j}} p^{-r}\right) = \mathfrak{N}_p\left(\sum_{j=1}^{n} \mathbf{d}_j\right) = \Sigma\left(\sum_{j=1}^{n} \underline{\mathbf{d}}^{(j)}\right).$$

Since

$$\begin{aligned}\Sigma\left(\sum_{j=1}^{n} \underline{\mathbf{d}}^{(j)}\right) = \Sigma\left(\tau\left(\sum_{j=1}^{n} \underline{\mathbf{d}}^{(j)}\right)\right) &= \Sigma\left(\tau\left(\sum_{j=1}^{n} \underline{e}^{(j)}\right)\right)\\ &\leq \Sigma\left(\sum_{j=1}^{n} \underline{e}^{(j)}\right) = \sum_{j=1}^{n} \Sigma\left(\underline{e}^{(j)}\right) \leq n \cdot \mathrm{dom}_p(M(\underline{A})),\end{aligned}$$

one concludes that $\Sigma\left(\underline{e}^{(j)}\right) = \mathrm{dom}_p(M(\underline{A}))$ for every $j = 1, \cdots, n$ and $\sum_{j=1}^{n} \underline{e}^{(j)} \in \mathbb{P}$. Consequently, one has $\sum_{j=1}^{n} \underline{e}^{(j)} = \sum_{j=1}^{n} \underline{\mathbf{d}}^{(j)}$, implying that $\sum_{j=1}^{n} e_j = \sum_{j=1}^{n} \mathbf{d}_j$.

Note that there is no duplication among $e_1, \cdots, e_n$: if not, then there exists a coordinate of $\sum_{j=1}^{n} \underline{e}^{(j)}$ that is at least 2, contradicting the fact that every coordinate of $\sum_{j=1}^{n} \underline{\mathbf{d}}^{(j)}$ is 0 or 1. If one writes $e_j = \sum_{r\in A_{k_j}} p^{-r}$ for every $j = 1, \cdots, n$, then

$$\sum_{r\in\bigsqcup_{j=1}^{n} A_{k_j}} p^{-r} = \sum_{j=1}^{n} e_j = \sum_{j=1}^{n} \mathbf{d}_j = \sum_{r\in\bigsqcup_{j=1}^{n} A_{i_j}} p^{-r},$$

implying that $\bigsqcup_{j=1}^{n} A_{k_j} = \bigsqcup_{j=1}^{n} A_{i_j}$. This forces $A_{k_j} = A_{i_j}$ and consequently $e_j = \mathbf{d}_j$ for every $j = 1, \cdots, n$, up to a permutation of $\{1, \cdots, n\}$. □

We end this section with the following technical lemma, which will be used in the proof of Theorem 1.7 to extract the orphaned exponents in the multinomial expansion.

Lemma 3.8. — *Let $S \subset \mathbb{P}$ be a (c, n)-sparse subset for some integers $c, n \geq 1$. Let*

$$\phi_0 \colon S \longrightarrow \mathbf{N},\ \underline{d} \longmapsto |\{i | \underline{d} = \underline{d}^{(i)}\}|.$$

Then ϕ_0 is the unique function $\phi \colon S \to \mathbf{N}$ such that $\sum_{\underline{d}\in S} \phi(\underline{d}) \leq n$ and

$$\sum_{\underline{d}\in S} \|\underline{d}\| \cdot \phi(\underline{d}) \equiv \sum_{\underline{d}\in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}.$$

Proof. — Suppose that $\phi_1 \colon S \to \mathbf{N}$ is another function such that $\sum_{\underline{d}\in S} \phi_1(\underline{d}) \leq n$ and

$$\sum_{\underline{d}\in S} \|\underline{d}\| \cdot \phi_1(\underline{d}) \equiv \sum_{\underline{d}\in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}.$$

1. If $\sum_{\underline{d}\in S} \phi_1(\underline{d}) < n$, then by Lemma 3.3 (4) we have

$$\Sigma\left(\tau\left(\sum_{\underline{d}\in S} \underline{d} \cdot \phi_1(\underline{d})\right)\right) \leq \Sigma\left(\sum_{\underline{d}\in S} \phi_1(\underline{d}) \cdot \underline{d}\right) = \sum_{\underline{d}\in S} \phi_1(\underline{d}) \sum_{i=1}^{\infty} \underline{d}_i < n \cdot c.$$

   On the other hand, the sparseness condition (2a) of Definition 3.5 implies that $\sum_{j=1}^{n} d_i^{(j)} < p$ for any i, hence

$$\sum_{\underline{d}\in S} \underline{d} \cdot \phi_0(\underline{d}) = \left(\sum_{j=1}^{n} d_i^{(j)}\right)_{i\in\mathbf{Z}_{\geq 0}} = \tau\left(\sum_{\underline{d}\in S} \underline{d} \cdot \phi_0(\underline{d})\right)$$

and consequently

$$\Sigma\left(\tau\left(\sum_{\underline{d}\in S}\underline{d}\cdot\phi_0(\underline{d})\right)\right)=\sum_{i=0}^{\infty}\left(\sum_{j=1}^{n}d_i^{(j)}\right)=\sum_{j=1}^{n}\Sigma(d^{(j)})=n\cdot c.$$

This leads to a contradiction by Lemma 3.3 (2).

2. If $\sum_{\underline{d}\in S}\phi_1(\underline{d})=n$, then we take $\underline{e}^{(1)},\underline{e}^{(2)},\cdots,\underline{e}^{(n)}\in S$ such that $\phi_1(\underline{d})=|\{i|\underline{d}=\underline{e}^{(i)}\}|$ for any $\underline{d}\in S$. Then

$$\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi_1(\underline{d})-\sum_{\underline{d}\in S}\|\underline{d}\|\cdot\phi_0(\underline{d})=\left\|\sum_{\underline{d}\in S}\underline{d}\cdot\phi_1(\underline{d})\right\|-\left\|\sum_{\underline{d}\in S}\underline{d}\cdot\phi_0(\underline{d})\right\|=\left\|\sum_{i=1}^{n}\underline{e}^{(i)}\right\|-\left\|\sum_{i=1}^{n}\underline{d}^{(i)}\right\|\in\mathbf{Z}.$$

By the sparseness condition (2b) of Definition 3.5, up to a permutation of $\{1,2,\cdots,n\}$, $\underline{d}^{(i)}=\underline{e}^{(i)}$ for any $i=1,2,\cdots,n$. This implies that $\phi_1=\phi_0$.

□

4. T-scaled realization of $\mathbf{L}_p$

To prove Theorem 1.7, one must group the terms of a p-adic Hahn series $f=\sum_{q\in\mathbf{Q}}[f(q)]p^q$ in $\mathbf{L}_p$ with $\mathrm{Supp}(f)\subset\frac{1}{T}\mathbf{Z}[1/p]$ by the residue of the exponent modulo $\frac{1}{T}\mathbf{Z}$ for some integer $T\geq 1$. This works when $T=1$, since a direct computation shows that the element

$$\sum_{q\in\mathrm{Supp}(f)/\mathbf{Z}}\left(\sum_{w\in\mathbf{Z}}f(q+w)p^w\right)t^q\in W(\overline{\mathbf{F}}_p)(\!(t^{\mathbf{Q}})\!)$$

is a preimage of f under the natural projection $W(\overline{\mathbf{F}}_p)(\!(t^{\mathbf{Q}})\!)\to W(\overline{\mathbf{F}}_p)(\!(p^{\mathbf{Q}})\!)$, where

$$\mathrm{Supp}(f)/\mathbf{Z}:=\{\inf(\mathrm{Supp}(f)\cap(q+\mathbf{Z}))|q\in\mathrm{Supp}(f)\}\subset\mathbf{Q}$$

is a well-ordered set of representatives of $\mathrm{Supp}(f)$ modulo $\mathbf{Z}$. However, when $T>1$, the same construction fails, because the element $\sum_{w\in\frac{1}{T}\mathbf{Z}}f(q+w)p^w$ does not necessarily lie in $W(\overline{\mathbf{F}}_p)$, so a direct analogue of the above construction is not well-defined(3). To resolve this, we enlarge the ring of Witt vectors $W(\overline{\mathbf{F}}_p)$ to include $p^{1/T}$, and realize $\mathbf{L}_p$ as a quotient of $W(\overline{\mathbf{F}}_p)[p^{1/T}](\!(t^{\mathbf{Q}})\!)$ by a suitable ideal. This is the main content of this section.

Throughout the rest of this section, T denotes a positive integer.

lem:61648 ***Lemma 4.1.*** — *One has* $\left[\breve{\mathbf{Q}}_p(p^{1/T}):\breve{\mathbf{Q}}_p\right]=T$. *In particular,* $1,p^{1/T},\cdots,p^{(T-1)/T}$ *form a basis of* $\breve{\mathbf{Q}}_p(p^{1/T})$ *over* $\breve{\mathbf{Q}}_p$.

Proof. — Since $\breve{\mathbf{Q}}_p$ is a discrete valuation field and $p^{1/T}$ is a root of the Eisenstein polynomial X^T-p, the result follows from the Eisenstein criterion. □

Remark 4.2. — *$W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$ is a subfield of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ via the natural inclusion $W(\overline{\mathbf{F}}_p)\subset W(\overline{\mathbf{F}}_p)[p^{1/T}]$.*

Definition 4.3. — *Let $\mathcal{N}_T$ be the set of elements $\sum_{q\in\mathbf{Q}}c_qt^q\in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ such that for any $q\in\mathbf{Q}$,*

$$\sum_{n\in\mathbf{Z}}c_{q+\frac{n}{T}}p^{\frac{n}{T}}=0.$$

*We call these elements the T-**null-series** in $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.*

---

(3)This issue was detected during the formalization of the proof of the main theorem in Lean 4.

Remark 4.4. — *When $T = 1$, $\mathcal{N}_T$ coincides with the null-series in $W(\overline{\mathbf{F}}_p)(\!(p^{\mathbf{Q}})\!)$ defined in Proposition 2.3.*

Lemma 4.5. —

1. *$\mathcal{N}_T$ is an ideal of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.*
2. *For every element f of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, there exists a unique element $g = \sum_{q\in\mathbf{Q}}[g(q)]t^q$ in $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ such that $f - g \in \mathcal{N}_T$.*
3. *$\mathcal{N}_T$ is a maximal ideal of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, so that $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$ is a field.*

Proof. — These three statements are generalizations of [Poo93, Proposition 3], [Poo93, Proposition 4] and [Poo93, Corollary 3] respectively, and the proofs are essentially the same. □

Remark 4.6. — *By (2) of this lemma, we will formally write elements of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$ as $\sum_{q\in\mathbf{Q}}[g(q)]p^q$.*

Lemma 4.7. — *One has $\mathcal{N}_T \cap W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) = \mathcal{N}$.*

Proof. — Let $f = \sum_{q\in\mathbf{Q}}[f(q)]t^q \in \mathcal{N}$, then for any $q \in \mathbf{Q}$,

eq:64661 (a)
$$\sum_{n\in\mathbf{Z}}[f(q+n)]p^n = 0.$$

Notice that for any $q \in \mathbf{Q}$, one has

$$\begin{aligned}\sum_{n\in\mathbf{Z}}\left[f\left(q+\frac{n}{T}\right)\right]p^{\frac{n}{T}} &= \sum_{u=0}^{T-1}\sum_{\substack{n\in\mathbf{Z}\\ n\equiv u \pmod T}}\left[f\left(q+\frac{n}{T}\right)\right]p^{\frac{n}{T}}\\ &= \sum_{u=0}^{T-1}p^{\frac{u}{T}}\sum_{n\in\mathbf{Z}}\left[f\left(\left(q+\frac{u}{T}\right)+n\right)\right]p^n.\end{aligned}$$

By applying (a) to $q + \frac{u}{T}$ for $u = 0, 1, \cdots, T-1$, one has $\sum_{n\in\mathbf{Z}}\left[f\left(q+\frac{n}{T}\right)\right]p^{\frac{n}{T}} = 0$. As a result, $f \in \mathcal{N}_T$.

Conversely, let $g = \sum_{q\in\mathbf{Q}}[g(q)]t^q \in \mathcal{N}_T$. Then for any $q \in \mathbf{Q}$, one has

$$\sum_{n\in\mathbf{Z}}\left[g\left(q+\frac{n}{T}\right)\right]p^{\frac{n}{T}} = \sum_{u=0}^{T-1}p^{\frac{u}{T}}\sum_{n\in\mathbf{Z}}\left[g\left(\left(q+\frac{u}{T}\right)+n\right)\right]p^n = 0.$$

By Lemma 4.1, for any $u = 0, 1, \cdots, T-1$, one has $\sum_{n\in\mathbf{Z}}\left[g\left(\left(q+\frac{u}{T}\right)+n\right)\right]p^n = 0$. The result follows from taking $u = 0$. □

Lemma 4.8. — *One has $W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) + \mathcal{N}_T = W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.*

Proof. — Take $f = \sum_{q\in\mathbf{Q}} c_q t^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, where

$$c_q = \sum_{n=0}^{\infty}[c_{q,n}]p^{\frac{n}{T}} \in W(\overline{\mathbf{F}}_p)[p^{1/T}].$$

Then one has $f = f_0 + p^{\frac{1}{T}}f_1 + \cdots + p^{\frac{T-1}{T}}f_{T-1}$, where

$$f_u = \sum_{t\in\mathbf{Q}}\left(\sum_{\substack{n\in\mathbf{N}\\ n\equiv u \pmod T}}[c_{q,n}]p^{\frac{n-u}{T}}\right)t^q \in W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})),\ u = 0, 1, \cdots, T-1.$$

It suffices to show that $p^{\frac{u}{T}} f_u$ belongs to $W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) + \mathcal{N}_T$ for $u = 0, 1, \cdots, T-1$. This is clear if we write f_u as $f_u = (p^{\frac{u}{T}} \cdot t^0 - t^{\frac{u}{T}}) f_u + t^{\frac{u}{T}} f_u$, where $t^{\frac{u}{T}} f_u \in W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$ and $(p^{\frac{u}{T}} \cdot t^0 - t^{\frac{u}{T}}) f_u$ is a T-null-series. □

Proposition 4.9. — *There is an isomorphism of valued fields*

$$\sigma\colon \mathbf{L}_p \xrightarrow{\cong} W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T.$$

Moreover, the following diagram commutes:

eq:28959 (b)
$$\begin{tikzcd} W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}})) \arrow[r, hook, "\iota"] \arrow[d, two heads, "\pi"] & W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}})) \arrow[d, two heads, "\pi_T"] \\ \mathbf{L}_p \arrow[r, "\cong", "\sigma"'] & W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T \end{tikzcd}.$$

Proof. — This is a direct consequence of the following standard fact in commutative algebra: given domains $R_1 \subset R_2$ and an ideal I of R_2, if $R_1 + I = R_2$, then $R_1/(I \cap R_1) \cong R_2/I$, where the isomorphism is induced by the inclusion $R_1 \subset R_2$. □

Remark 4.10. — *For any element $f = \sum_{q\in\mathbf{Q}}[f(q)]p^q \in \mathbf{L}_p$, its lift $\widehat{f} = \sum_{q\in\mathbf{Q}}[f(q)]t^q \in W(\overline{\mathbf{F}}_p)((t^{\mathbf{Q}}))$ embeds into $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$ and then projects to an element of $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$, which can also be formally written as $\sum_{q\in\mathbf{Q}}[f(q)]p^q$. The commutative diagram in the proposition above shows that the isomorphism between $\mathbf{L}_p$ and $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$ is given by*

$$\sum_{q\in\mathbf{Q}}[f(q)]p^q \in \mathbf{L}_p \longmapsto \sum_{q\in\mathbf{Q}}[f(q)]p^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T.$$

5. Main theorem and its applications

5.1. Main theorem. —

thm:23791 ***Theorem 5.1.*** — *Let $f \in \mathbf{L}_p$ be a p-adic Hahn series such that there exists an integer $T \geq 1$ for which $-T \cdot \mathrm{Supp}(f)$ admits a sparse set $W \neq 0$ of representatives modulo $\mathbf{Z}$. Then f is transcendental over $\breve{\mathbf{Q}}_p$, and hence over $\mathbf{Q}_p$.*

Proof. — We will use the notation of (b) freely throughout the proof without further explanation.

Suppose that f is a root of a polynomial $P(X) = a_n X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in W(\overline{\mathbf{F}}_p)[X]$ with $a_n \neq 0$. By multiplying $P(X)$ by a suitable power of X, we may assume that there exists $S \subset \mathbb{P}$ such that $\|S\| \coloneqq \{\|\underline{d}\| \in \mathbf{Q} | \underline{d} \in S\}$ is a set of representatives of $-T_f \cdot \mathrm{Supp}(f)$ modulo $\mathbf{Z}$, and S is (c, n)-sparse for some integer $c \geq 1$ (cf. Lemma 3.6), where n equals the degree of $P(X)$. Thus there exist n elements $\underline{d}^{(1)}, \underline{d}^{(2)}, \cdots, \underline{d}^{(n)} \in S$ satisfying condition (2) of Definition 3.5 for n and c.

Since $\|S\|$ is a set of representatives of $-T_f \cdot \mathrm{Supp}(f)$ modulo $\mathbf{Z}$, the set $-\frac{1}{T}\|S\|$ is a set of representatives of $\mathrm{Supp}(f)$ modulo $\frac{1}{T_f}\mathbf{Z}$. For every $\underline{d} \in S$, we set

$$n_{\underline{d}} \coloneqq -\frac{\|\underline{d}\|}{T} - \inf\Big(\mathrm{Supp}(f) \cap \Big(-\frac{\|\underline{d}\|}{T} + \frac{1}{T}\mathbf{Z}\Big)\Big) \in \frac{1}{T}\mathbf{Z}.$$

This element is well defined because $\mathrm{Supp}(f) \cap \Big(-\frac{\|\underline{d}\|}{T} + \frac{1}{T}\mathbf{Z}\Big)$ is non-empty by the assumption that $-\frac{1}{T}\|S\|$ is a set of representatives of $\mathrm{Supp}(f)$ modulo $\frac{1}{T}\mathbf{Z}$, and is a subset of the well-ordered set $\mathrm{Supp}(f)$, and hence has a minimum element. Then the set

$$\widetilde{S} \coloneqq \Big\{-\frac{\|\underline{d}\|}{T} - n_{\underline{d}} \Big| \underline{d} \in S\Big\} \subseteq \mathrm{Supp}(f)$$

is a well-ordered set of representatives of $\mathrm{Supp}(f)$ modulo $\frac{1}{T}\mathbf{Z}$, and is in bijection with S via the map

$$\mu\colon \underline{d} \longmapsto -\frac{\|\underline{d}\|}{T} - n_{\underline{d}} = \inf\left(\mathrm{Supp}(f) \cap \left(-\frac{\|\underline{d}\|}{T} + \frac{1}{T}\mathbf{Z}\right)\right).$$

Suppose that $f = \sum_{q\in\mathbf{Q}}[f(q)]p^q \in \mathbf{L}_p$. Then $\sigma(f) = \sum_{q\in\mathbf{Q}}[f(q)]p^q \in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))/\mathcal{N}_T$, and clearly $\mathrm{Supp}(\sigma(f)) = \mathrm{Supp}(f)$. For any $s \in \widetilde{S}$, set $C_s := \sum_{w\in\mathbf{Z}}[f(s+\frac{w}{T})]p^{\frac{w}{T}} \in \breve{\mathbf{Q}}_p(p^{1/T})$. By the construction of $\widetilde{S}$, we have $C_s \neq 0$ and $f(s+\frac{w}{T}) = 0$ for every $s \in \widetilde{S}$ and every integer $w < 0$. Thus $C_s = \sum_{w=0}^{\infty}[f(s+\frac{w}{T})]p^{\frac{w}{T}} \in W(\overline{\mathbf{F}}_p)[p^{1/T}]$, so that the element

$$\widehat{f} := \sum_{s\in\widetilde{S}} C_s \cdot t^s \in W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$$

is a lift of $\sigma(f)$ in $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$.

By multinomial expansion, for any integer $i \geq 0$, one has

$$\widehat{f}^i = \sum_{q\in\mathbf{Q}}\left(\sum_{\substack{s_1,\cdots,s_i\in\widetilde{S}\\ s_1+\cdots+s_i=q}} \prod_{k=1}^{i} C_{s_k}\right)t^q = \sum_{q\in\mathbf{Q}}\left(\sum_{\substack{\widetilde{\phi}\colon \widetilde{S}\to\mathbf{N}\\ \sum_{s\in\widetilde{S}}\widetilde{\phi}(s)=i\\ \sum_{s\in\widetilde{S}}\widetilde{\phi}(s)\cdot s=q}} \frac{i!}{\prod_{s\in\widetilde{S}}\widetilde{\phi}(s)!}\prod_{s\in\widetilde{S}} C_s^{\widetilde{\phi}(s)}\right)t^q.$$

Since

$$\sigma\left(\sum_{i=0}^{n} a_i f^i\right) = \sum_{i=0}^{n} a_i\sigma(f)^i = 0$$

the element

$$\begin{aligned}\sum_{i=0}^{n} a_i\widehat{f}^i &= \sum_{q\in\mathbf{Q}}\left(\sum_{i=0}^{n} a_i \sum_{\substack{\widetilde{\phi}\colon \widetilde{S}\to\mathbf{N}\\ \sum_{s\in\widetilde{S}}\widetilde{\phi}(s)=i\\ \sum_{s\in\widetilde{S}}\widetilde{\phi}(s)\cdot s=q}} \frac{i!}{\prod_{s\in\widetilde{S}}\widetilde{\phi}(s)!}\prod_{s\in\widetilde{S}} C_s^{\widetilde{\phi}(s)}\right)t^q\\ &= \sum_{q\in\mathbf{Q}}\left(\sum_{\substack{\widetilde{\phi}\colon \widetilde{S}\to\mathbf{N}\\ \sum_{s\in\widetilde{S}}\widetilde{\phi}(s)\leq n\\ \sum_{s\in\widetilde{S}}\widetilde{\phi}(s)\cdot s=q}} a_{\sum_{s\in\widetilde{S}}\widetilde{\phi}(s)} \frac{\left(\sum_{s\in\widetilde{S}}\widetilde{\phi}(s)\right)!}{\prod_{s\in\widetilde{S}}\widetilde{\phi}(s)!}\prod_{s\in\widetilde{S}} C_s^{\widetilde{\phi}(s)}\right)t^q\end{aligned}$$

is a T-null-series in $W(\overline{\mathbf{F}}_p)[p^{1/T}]((t^{\mathbf{Q}}))$, i.e.

$$\sum_{w\in\mathbf{Z}} p^w \sum_{\substack{\widetilde{\phi}\colon \widetilde{S}\to\mathbf{N}\\ \sum_{s\in\widetilde{S}}\widetilde{\phi}(s)\leq n\\ \sum_{s\in\widetilde{S}}\widetilde{\phi}(s)\cdot s=q+\frac{w}{T}}} a_{\sum_{s\in\widetilde{S}}\widetilde{\phi}(s)} \frac{\left(\sum_{s\in\widetilde{S}}\widetilde{\phi}(s)\right)!}{\prod_{s\in\widetilde{S}}\widetilde{\phi}(s)!}\prod_{s\in\widetilde{S}} C_s^{\widetilde{\phi}(s)} = 0$$

for every $q \in \mathbf{Q}$. We specialize this at $q = -\frac{r_0}{T}$, where

$$r_0 \coloneqq \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}),$$

to get

eq:37911 (c)
$$\sum_{w \in \mathbf{Z}} p^w \sum_{\substack{\widetilde{\phi}\colon \widetilde{S} \to \mathbf{N} \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \leq n \\ \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s = -\frac{r_0}{T} + \frac{w}{T}}} a_{\sum_{s \in \widetilde{S}} \widetilde{\phi}(s)} \frac{\left(\sum_{s \in \widetilde{S}} \widetilde{\phi}(s)\right)!}{\prod_{s \in \widetilde{S}} \widetilde{\phi}(s)!} \prod_{s \in \widetilde{S}} C_s^{\widetilde{\phi}(s)} = 0.$$

For every $\widetilde{\phi}\colon \widetilde{S} \to \mathbf{N}$ such that $\sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \leq n$ and $\sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s \equiv -\frac{r_0}{T} \pmod{\frac{1}{T}\mathbf{Z}}$, we set $\phi \coloneqq \widetilde{\phi} \circ \mu$. Then

$$\sum_{\underline{d} \in S} \phi(\underline{d}) = \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \leq n$$

and

$$\begin{aligned} \sum_{\underline{d} \in S} \phi(\underline{d}) \cdot \|\underline{d}\| &= -T \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s - \sum_{\underline{d} \in S} \widetilde{\phi}\left(-\frac{\|\underline{d}\|}{T} - n_{\underline{d}}\right)(T \cdot n_{\underline{d}}) \\ &\equiv -T \sum_{s \in \widetilde{S}} \widetilde{\phi}(s) \cdot s \equiv r_0 \pmod{\mathbf{Z}}. \end{aligned}$$

By the claim, one knows that $\phi = \phi_0$, i.e. $\widetilde{\phi} = \phi_0 \circ \mu^{-1}$. As a result, one can rewrite (c) as

$$p^{T \sum_{\underline{d} \in S} \phi_0(\underline{d}) \cdot \mu(\underline{d}) + r_0} a_n \frac{n!}{\prod_{\underline{d} \in S} \phi_0(\underline{d})!} \prod_{\underline{d} \in S} C_{\mu(\underline{d})}^{\phi_0(\underline{d})} = 0,$$

implying that $a_n = 0$ or $C_{\mu(\underline{d})} = 0$ for some $\underline{d} \in S$. Both cases lead to contradictions. The result follows. $\square$

Remark 5.2. — *In the above proof, the blue sums are in fact finite by [Poo93, Lemma 1], while the red sums and products are well-defined because the function inside is finitely supported by the definition of $\widetilde{\phi}$.*

5.2. Applications. — A typical application of Theorem 1.7 is to show that if the decimal parts of any two elements of the sign-inverted support $-\operatorname{Supp}(f)$ of a p-adic Hahn series f have non-overlapping base-p digits, then f is transcendental over $\check{\mathbf{Q}}_p$:

prop:15994 ***Proposition 5.3.*** — *Let $\underline{A} \coloneqq (A_i)_{i \geq 1}$ be a family of pair-wise disjoint nonempty subsets of $\mathbf{N}$ such that $\sup_i |A_i| < \infty$. If the set of the form*

$$M_c(\underline{A}) \coloneqq \left\{ \frac{1}{T}\left(c_i - \sum_{r \in A_i} p^{-r}\right) \middle| c_i \in \mathbf{Z},\ i = 1, 2, \cdots \right\},\ T \in \mathbf{Z}_{\geq 1}$$

is well-ordered, then any p-adic Hahn series with support $M_c(\underline{A})$ is transcendental over $\check{\mathbf{Q}}_p$.

Proof. — Let $f \in \mathbf{L}_p$ be a p-adic Hahn series with support $M_c(\underline{A})$. Let

$$\lambda\colon M(\underline{A}) \longrightarrow M_c(\underline{A}),\ \sum_{r \in A_i} p^{-r} \longmapsto \frac{1}{T}\left(c_i - \sum_{r \in A_i} p^{-r}\right),$$

which is obviously a bijection.

Since there is at most one A_i that contains 0, and removing finitely many terms from a p-adic Hahn series does not change its algebraicity, we may assume that $0 \notin A_i$ for every $i \geq 1$.

Let $C \in \mathbf{Z}_{\geq 1}$ be the maximal integer such that $|A_i| = C$ is attained for infinitely many i, and let $\underline{A}' := \{A_i \mid i = 1, 2, \cdots, \ |A_i| = C\}$. Since $C \leq \sup_i |A_i| < \infty$, the set $\underline{A} \backslash \underline{A}'$ is finite. If we set

$$M(\underline{A}') := \left\{ \sum_{r \in A_i} p^{-r} \middle| A_i \in \underline{A}' \right\},$$

Then $M(\underline{A}')$ is a sparse set by Example 3.7, and $M(\underline{A}) \backslash M(\underline{A}')$ is a finite set. Note that $\lambda(M(\underline{A}'))$ is a well-ordered subset of $\mathbf{Q}$, we can set

$$f' := \sum_{q \in \lambda(M(\underline{A}'))} [f(q)] p^q,$$

which is a well-defined element of $\mathbf{L}_p$. By Theorem 1.7, f' is transcendental over $\breve{\mathbf{Q}}_p$. Since the support of $f - f'$ is finite, f is also transcendental over $\breve{\mathbf{Q}}_p$. $\square$

As a simple corollary of Proposition 5.3, we prove the following mixed-characteristic analogue of the result of Huang and Ştefănescu (cf. Theorem 1.9):

Corollary 5.4. — *Let $f = \sum_{i=1}^{\infty} [f(i)] \cdot p^{-1/p^i} \in \mathbf{L}_p$ be a p-adic Hahn series. Then the following are equivalent:*

1. *the series f is algebraic over $\mathbf{Q}_p$;*
2. *the series f is algebraic over $\breve{\mathbf{Q}}_p$;*
3. *$f(i) = 0$ for all but finitely many i.*

Proof. — One only need to verify (2) implies (3).

If $\mathrm{Supp}(f)$ is infinite, then we may write $\mathrm{Supp}(f) = \{q_1, \cdots, \}$, where $q_i = -1/p^{k_i}$ for some integer $k_i \in \mathbf{Z}_{\geq 1}$. The result follows from taking $A_i = \{k_i\}$, $c_i = 0$ for every $i \geq 1$ and $T = 1$ in Proposition 5.3. $\square$

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